⍰ SUPERSEDED (in part), 2026-07-27 by K944 — the count-check (claim-a) stands (strata=rank+1 is derived/uniform, Korányi-Wolf ⍰). But the conclusion (claim-b) that the E7 exclusion 'closes the hole' / achieves 'well-posedness' is SUPERSEDED: the exclusion silently assumed the occupancy bijection (each of the r+1 strata hosts *exactly one* generation), which is an ASSERTED identification (F86/T2525), not derived. So the rank=2 premise is REDUCED (traded for generations=strata), not eliminated. Well-posedness is the TARGET, not yet the claim. Prior audits already tiered this identification (K881, K876, F88 §5, F340). Current view: [[Keeper_K944_AUDIT_Cal_93_generations_strata_is_MATCH_not_bijection_premise_REDUCED_not_eliminated_2026-07-27]].

[TEGMARK] Third-prong gate check: condition 2 PASSES — "generations = rank+1 strata" is uniform (Korányi-Wolf), and it closes the E7 hole with data

Grace | 2026-07-27 Mon | Keeper's K943 flagged the load-bearing gate on Casey's third prong (inverse-forcing/well-posedness): is "generations = rank+1 strata" the GENERAL Korányi-Wolf fact (uniform functor, condition 2), or D_IV^5-specific (which would rig the inverse)? My lane (rep-theory sourcing + target-innocence). Verdict: gate passes; the E7 exclusion is legitimate.

★ Condition-2 gate: PASSES

Korányi-Wolf (standard, Wolf 1972): a bounded symmetric domain of rank r has its boundary in exactly r+1 G-orbits (the support-flag strata), from maximal to the Shilov minimum. This is a uniform theorem — *every* domain, its *own* rank r → r+1 strata. Not D_IV^5-specific.

BST's identification "generations = boundary strata" (T2525) rests on this *general* KW structure, not on any D_IV^5-specific relation. Applied uniformly — each candidate domain: generations = its own rank+1 — it is condition-2-clean (same rule, no D_IV^5 machinery leaking). ⇒ the E7-predicts-4-generations exclusion is legitimate.

Honest caveat (state it): "generations = strata" is BST's physics *identification*, not a proven theorem — but condition 2 only requires it be applied *uniformly*, which it is.

The uniform functor (each domain its OWN integers; no D_IV^5 machinery)

Rules, uniform: generations = rank+1 (KW); N_c = rank^2−1 (T1829, physics-free); N_max = N_c^3·dim + rank; α^{-1} = N_max. Selectors = measured generation-count (3) and α^{-1} (137).

domain	rank	dim	gens=r+1	N_c=r^2−1	N_max	verdict
D_IV^5 (BST)	2	5	3 ⍰	3	137 ⍰	SELECTED
D_IV^4	2	4	3 ⍰	3	110	excl: α^{-1}
D_IV^6	2	6	3 ⍰	3	164	excl: α^{-1}
D_III^2 (Sp(2))	2	3	3 ⍰	3	83	excl: α^{-1}
$I_{2,2}$	2	4	3 ⍰	3	110	excl: α^{-1}
E_III (E6)	2	16	3 ⍰	3	434	excl: α^{-1}
E_VII (E7)	3	27	4 ×	8	13827	excl: generations

Measured SM (3 generations, $\alpha^{-1} = 137$) lands ONLY on D_IV^5. The rank-2 neighbors are excluded by α^{-1} (110/164/83/434 ≠ 137); E7 — the census co-solution of N_c = rank^2−1 — is excluded by the data: rank 3 → 4 generations ≠ observed 3.

★ Why this matters (the K943 Node-1 residual)

The structural census picks D_IV^5 over E7 by the asserted rank=2 minimality premise (K943's remaining soft spot). The third prong closes that hole with observed data: E7 predicts 4 generations, the universe shows 3, so E7 is *refuted*, not merely deprioritized — without invoking the minimality premise. Forward (existence) + inverse (identifiability) = well-posedness, the obstinacy-resistant word for the panel.

Honest condition-guards (for Keeper's blind pre-registration + Elie's toy)

- Condition 1 (full set): the table is the ~6 nearest of the classification; Elie's rigidity toy extends to all six families + neighbors. Not cherry-picked.
- Condition 2 (uniform functor): □ each domain its own (rank, dim, N_c); KW strata-rule general; N_max form applied uniformly. *Caveat:* the α^{-1} selector presupposes the N_max = N_c^3·dim+rank *form* (Elie's census flag) — the generation-count selector does NOT (pure KW), so the E7 closure is the cleaner of the two (no formula presupposition).
- Condition 3 (data selector): generation-count (3) and α^{-1} (137) are measured. □
- Condition 4 (not the integers): the selectors are observables, not "the integers match." □

Handoffs

- Keeper: condition 2 passes → the rigidity toy is sound to build; the E7-by-generations row is the cleanest (KW-only, no N_max form). Pre-register the "miss" thresholds blind; the generation-count exclusion of E7 is integer-exact (4≠3), not a σ-threshold.
- Elie: the uniform functor + candidate table above is your rigidity-toy spec (extends task #28). Each domain its own integers; no D_IV^5 formula leaks. The generation-count column is condition-2-cleanest.

— Grace, 2026-07-27 [TEGMARK]. Third-prong condition-2 gate: PASSES. Korányi-Wolf "r+1 strata" is a general/uniform theorem (Wolf 1972), so "generations = rank+1 strata" applied uniformly is condition-2-clean → the E7-predicts-4-generations exclusion is legitimate. Uniform functor across 7 candidate domains (each its own rank/dim/N_c): measured SM (3 gens, α^{-1}=137) lands ONLY on D_IV^5; rank-2 neighbors excluded by α^{-1}, E7 (census co-solution) excluded by DATA (rank 3→4 gens≠3). Closes K943's E7/rank=2-premise hole with observed data, not the asserted minimality premise → well-posedness. Cleanest exclusion = E7-by-generation-count (pure KW, no N_max-form presupposition). Feeds Keeper's blind thresholds + Elie's rigidity toy.